

Yiping Deng

Dublin, Ireland | yiping.deng.98@gmail.com | +353 85 720 7666 | linkedin.com/in/yipingdeng

Experience

Tech Lead II - AI Data, HubSpot | Dublin, Ireland Jun 2024 - Present

- Own and modernize Spark infrastructure; introduced AI-agent job auto-tuning, cutting Spark support load by 80%.
- Built cost tracking for a US\$6-7M/year Spark fleet; helped build HubSpot's Apache Iceberg infrastructure.
- Built an OpenTelemetry tracing and storage system covering all AI agents at HubSpot; use AI agents to analyze data-entity lineage graphs.
- Scaled vector infrastructure for RAG and semantic search to 7B+ production ANN vectors; current infrastructure supports 200+ indices.
- Built AI inference/feedback pipelines ingesting 2 TB/day; introduced JupyterHub and NVIDIA DGX H100 infrastructure.
- Grew the team to six people and helped establish two spin-out teams focused on JupyterHub and vector databases.

Tech Lead I - AI Data and MLOps, HubSpot | Dublin, Ireland Mar 2023 - Jun 2024

- Maintained a feature store aggregating internal APIs for AI model serving; implemented compliance workflows and scaled AI data collection across multiple data centres.

Senior Software Engineer I - AI Infrastructure, HubSpot | Dublin, Ireland Jan 2022 - Mar 2023

- Cut monthly AI inference and feedback storage costs from US\$90K to US\$5K by migrating away from HBase.

Software Engineer - AI Infrastructure, HubSpot | Dublin, Ireland Oct 2020 - Jan 2022

- Maintained Kubernetes GPU infrastructure serving 1B+ AI inferences/day and large-scale HBase storage for inference and feedback data.
- Standardized automated training pipelines for 6+ ML models, retrained weekly using user feedback.

Software Engineer Intern, HubSpot | Dublin, Ireland Sep 2019 - Aug 2020

- Implemented automated metrics and performance-tracking pipelines for ML models in the AI Infrastructure team.

Software Development Engineer Intern, Amazon | Luxembourg Jun 2019 - Aug 2019

- Built an internal UI for technical program managers to review machine-translated product catalogue content.

Research Assistant - Marine and Robotics Systems, Jacobs University Bremen Feb 2019 - May 2019

- Researched autonomous-driving algorithms using ROS, OpenCV and PyTorch.

Software Engineer, Working Student, Ubimax | Bremen, Germany Jun 2018 - May 2019

- Built a SqueezeNet training microservice for automotive use cases; deployed models to Google Glass Enterprise and Vuzix M300 with TensorFlow Lite.

Education

BSc Computer Science, Minor in Intelligent Systems, Jacobs University Bremen 2017 - 2020

- Coursework: machine learning, algorithms, data structures, operating systems and computer vision. Research in formal logic and marine robotics.

Publication

Hilbert Meets Isabelle: Formalisation of the DPRM Theorem in Isabelle May 2018

Co-author, EasyChair Preprint 152 | easychair.org/publications/preprint/GhvC

Skills and Open Source

Languages and ML: Python, Java, C++, Rust, Scala, CUDA, Triton, PyTorch, TensorFlow Lite

Data and infrastructure: Apache Spark, Apache Iceberg, Kubernetes, AWS, Snowflake, HBase, Qdrant, OpenTelemetry, JupyterHub

Areas: MLOps, LLMops, RAG, vector search, GPU programming, AI observability, data pipelines and lineage

Qdrant contributions: Four merged PRs covering peer mTLS, HTTPS client-certificate validation, live TLS certificate rotation and clustered-collection regression tests.